STAFF REVIEW · H446 108 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. 452.

2. 10.

3. A link/reference to the next node (for a singly linked list).

4. In its left subtree (under the chosen no-duplicate convention).

I Do — source worked steps

1. Model answer — After completing the current instruction and recognising the interrupt, the CPU preserves relevant state, runs the keyboard ISR to obtain the key event, restores state and resumes the interrupted program. — Check: link each claim to the scenario and use precise OCR terminology.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
The disk raises an interrupt; the CPU saves context when accepted; the ISR acknowledges completion and updates status; the original process resumes or the scheduler selects another ready process.

You Do — source mark schemes

1. Interrupt signals attention [2]; ISR is event-specific handler [2]; context is saved/restored to resume processing [2].

2. Normally checked at a suitable boundary, often after an instruction [2]; current state preserved [2]; ISR executed before next normal instruction and context restored [2].

3. Explain polling [2], event-driven interrupt [2], reduced wasted CPU checking [2], and interrupt overhead/priority or timing trade-off [2].

Exit Ticket — source answer

1. True. Relevant context is saved and restored.